

Learning to Trust

In-context trust-learning under a reputation vs. accuracy conflict

Question: when a source's STATED REPUTATION conflicts with its OBSERVED ACCURACY,
does a frozen-weight LLM learn to trust the accurate source — and how fast vs. a rational learner?

Model: claude-sonnet-4-6 (anthropic) | Source A = reputable but NOISIER, Source B = unestablished but ACCURATE

Accuracy gaps swept: $\sigma_A/\sigma_B = [1.5, 2.0, 3.0, 5.0]$ ($\sigma_B=12.0$) | Reputation strengths: ['faint', 'moderate', 'strong']

M=3 companies/round, T=16 rounds, ~20 games/condition | info-ladder: ['lean', 'self_history', 'summary', 'full']

10944 per-(round,company) observations | 2026-06-16

Method — per-round in-context loop

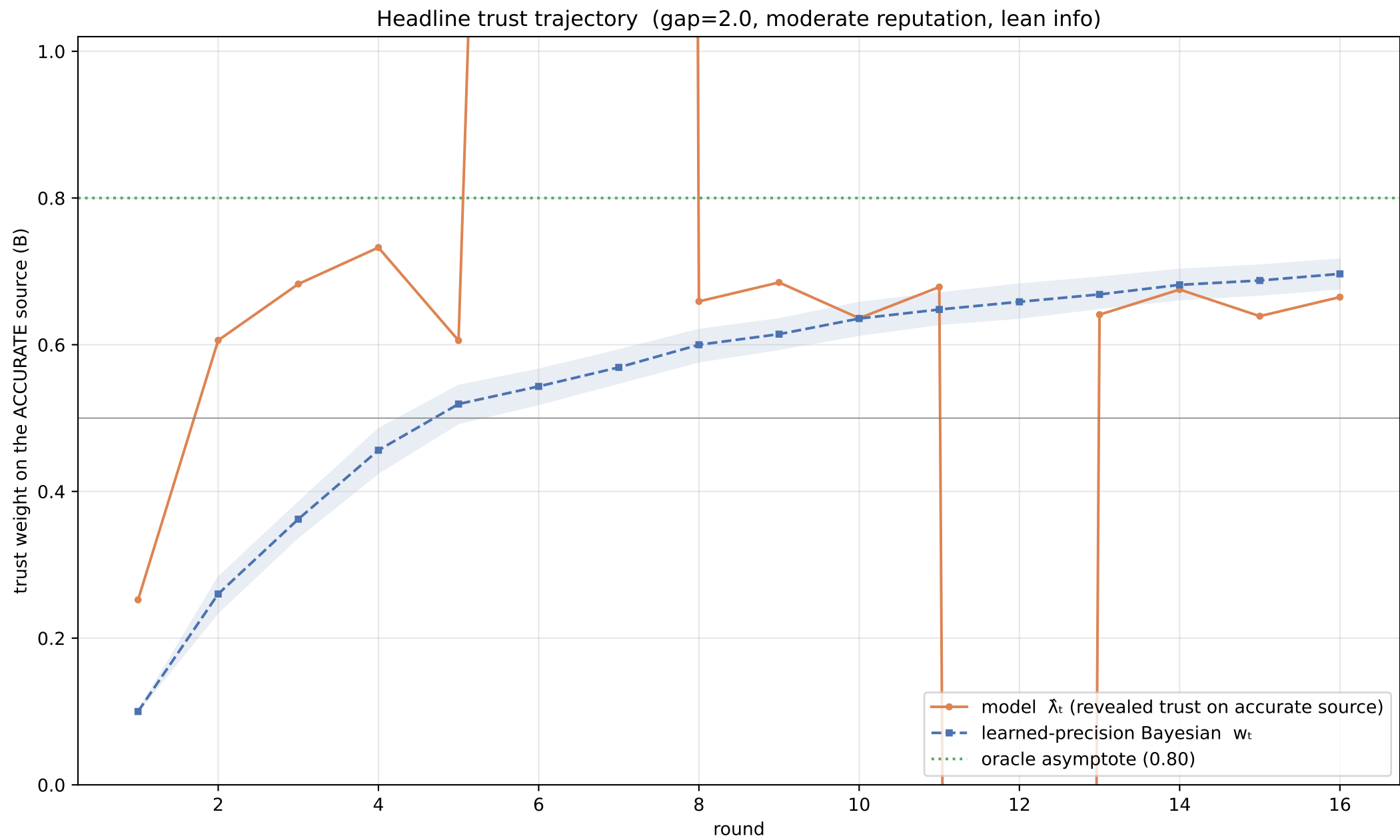
1. Build round-t prompt: persistent reputation header + RAW recap of past rounds (advisor record only) + this round's
2. Model outputs a numeric estimate per company (structured output) – the estimate IS the recommendation
3. Reveal the true value for each company -> accuracy evidence accumulates against the stale reputation
4. Extend history; proceed to round t+1 (one API call per round, T calls per game)

Reputation vs. accuracy CONFLICT: the header says A is trustworthy, but B's revealed errors are smaller every round.

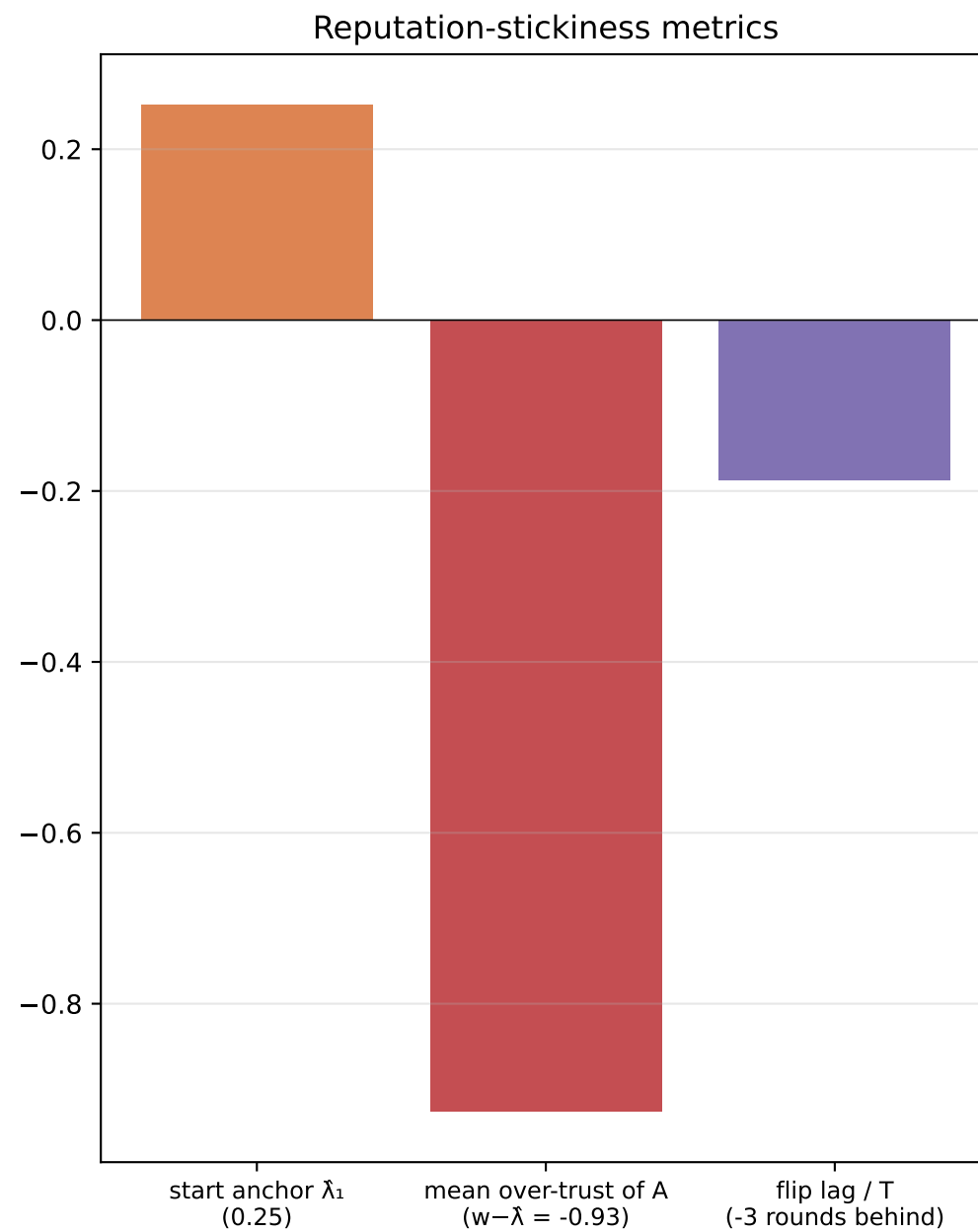
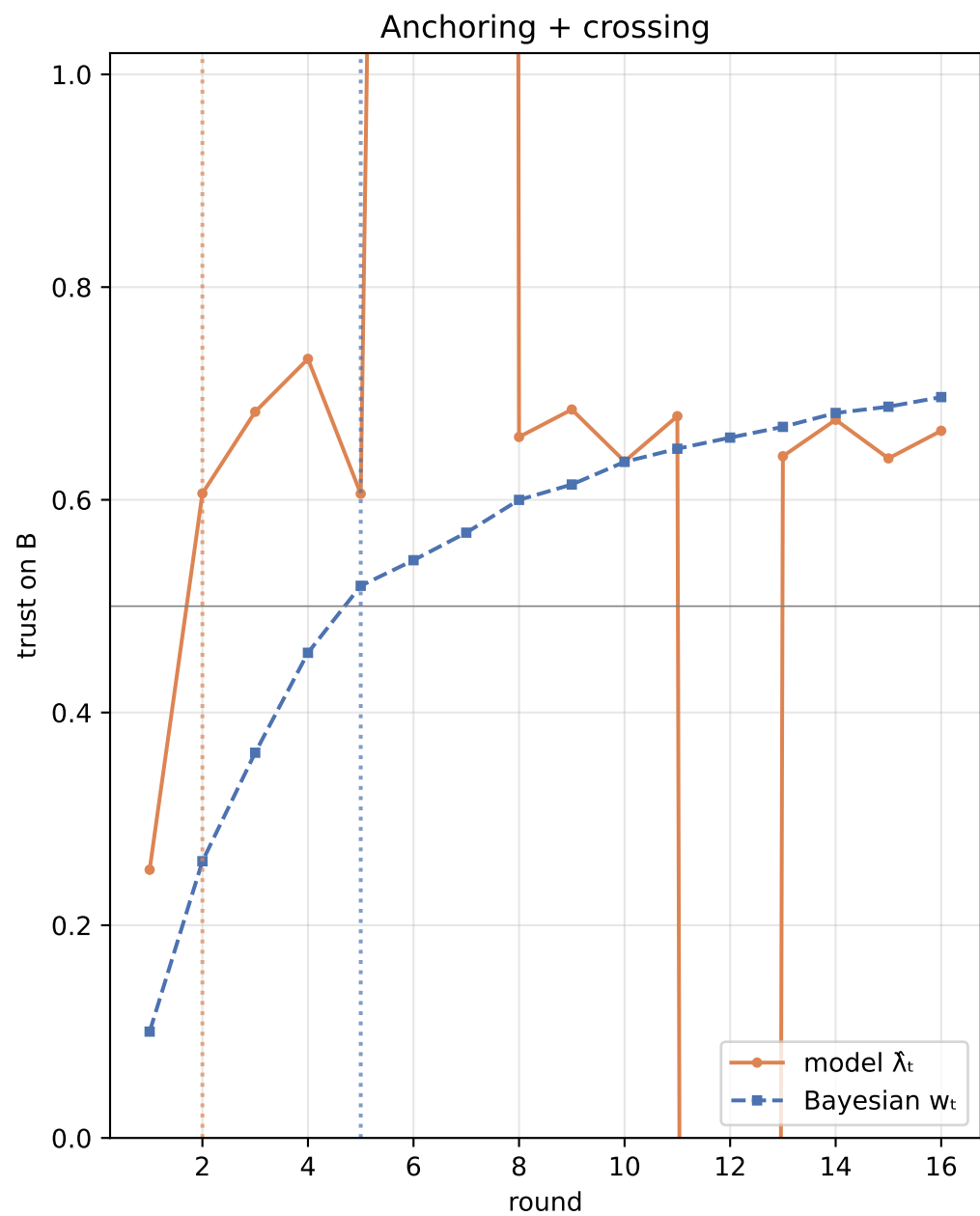
Normative baseline (LEARNED precision): a Bayesian that does NOT know the noise levels, starts from a reputation-derived prior favoring A, and updates each source's precision from realized errors: $\pi_{\hat{}} = (\nu_0 + n) / (\nu_0 \cdot \sigma^2 + S)$.

Trust on B: $w = \pi_{\hat{B}} / (\pi_{\hat{A}} + \pi_{\hat{B}})$ -> migrates A->B as evidence accrues (oracle asymptote uses the TRUE s

Model trust readout: per round t, regress $\text{model_est} \sim b_A \cdot a + b_B \cdot b$ across companies x games; $\lambda_{\hat{t}} = b_B / (b_A + b_B)$.

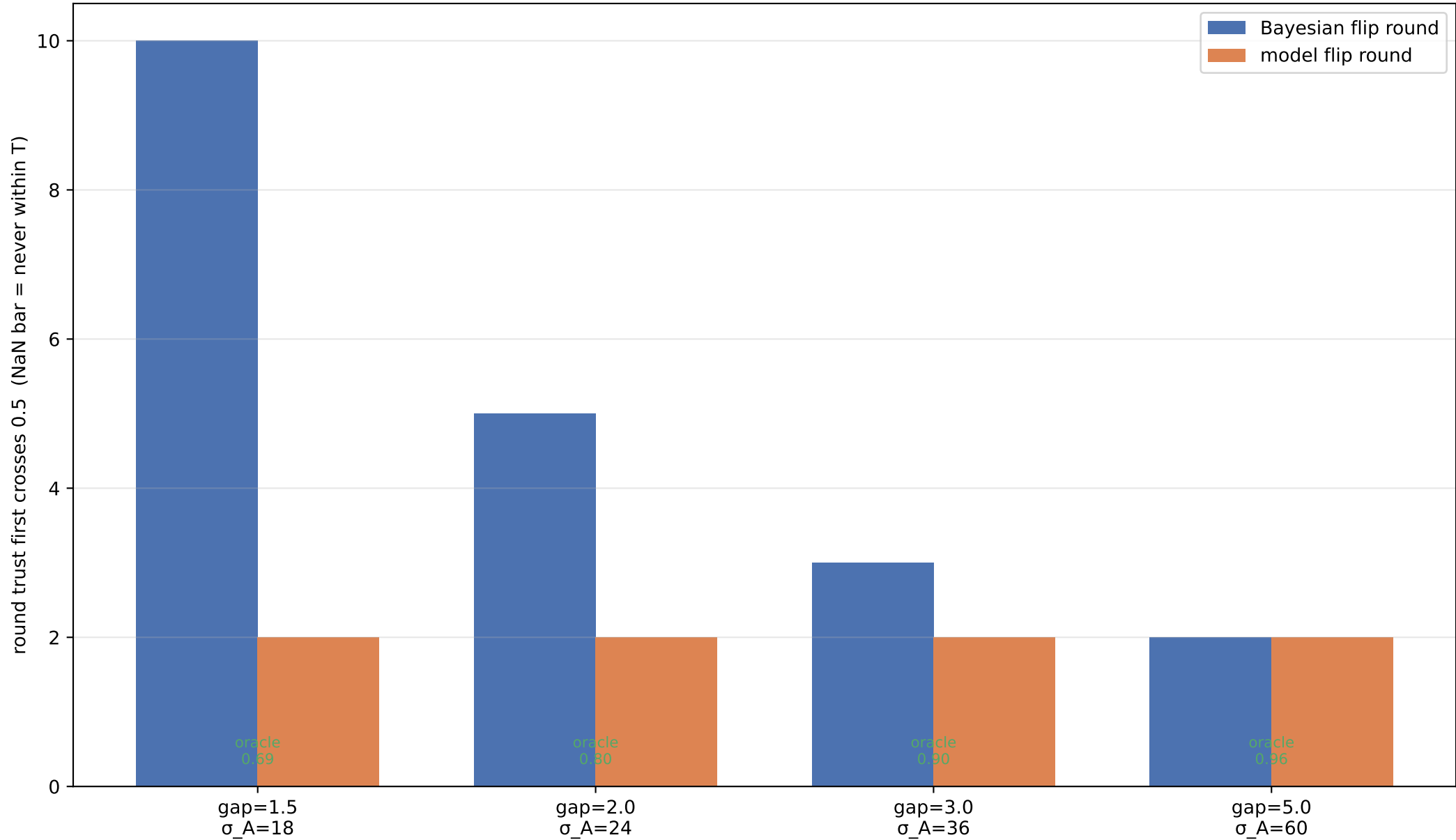


Both curves start anchored on the reputable-but-wrong source (low) and should climb toward the oracle. Model below Bayesian = reputation-stickiness; shaded = 95% bootstrap CI.



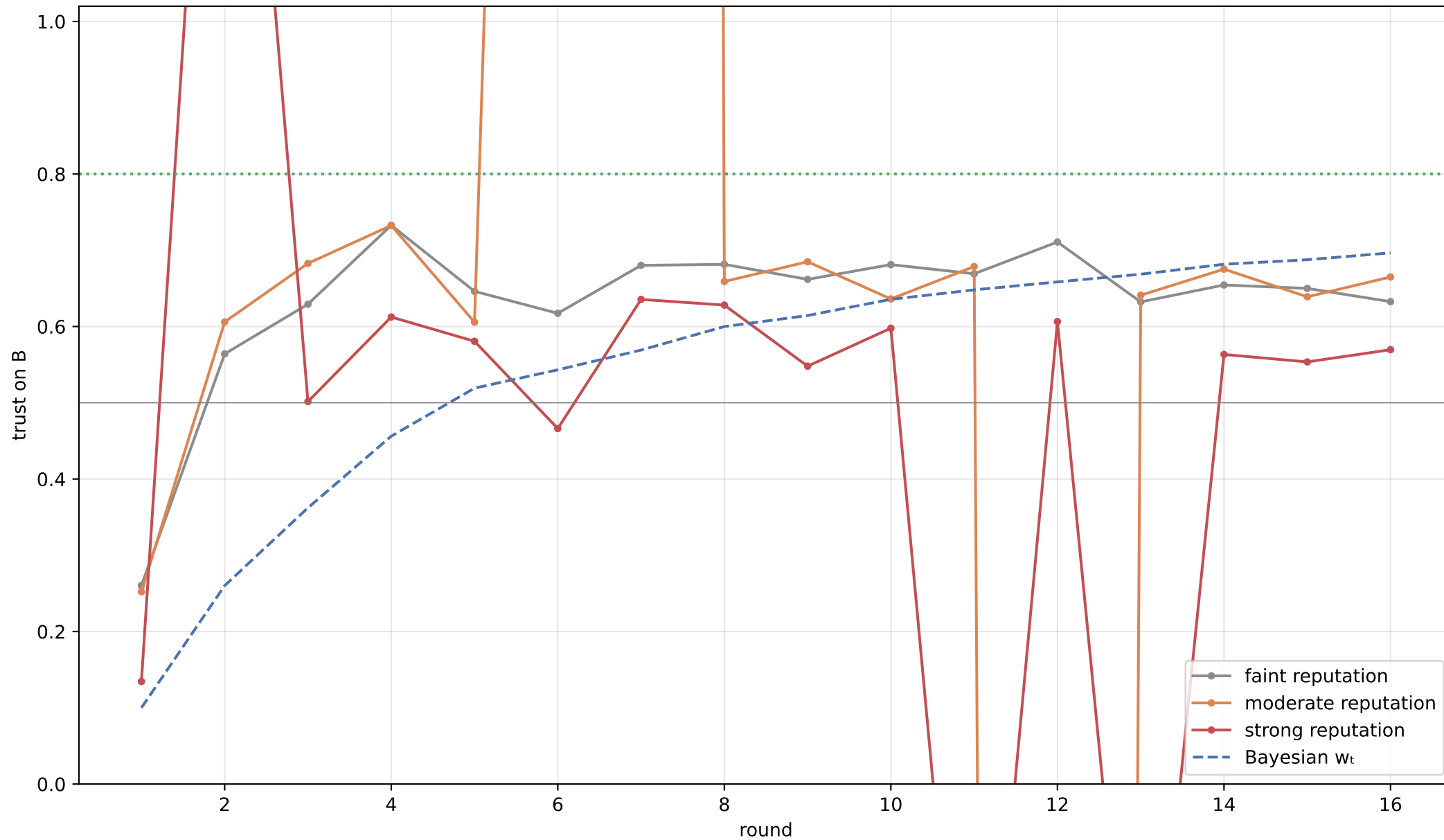
Stickiness = the model starts more anchored on A than warranted (left) and trails the rational A→B migration (positive over-trust, rounds-behind on the right).

DIFFICULTY (accuracy-gap) sweep — where trust transitions, model vs. Bayes



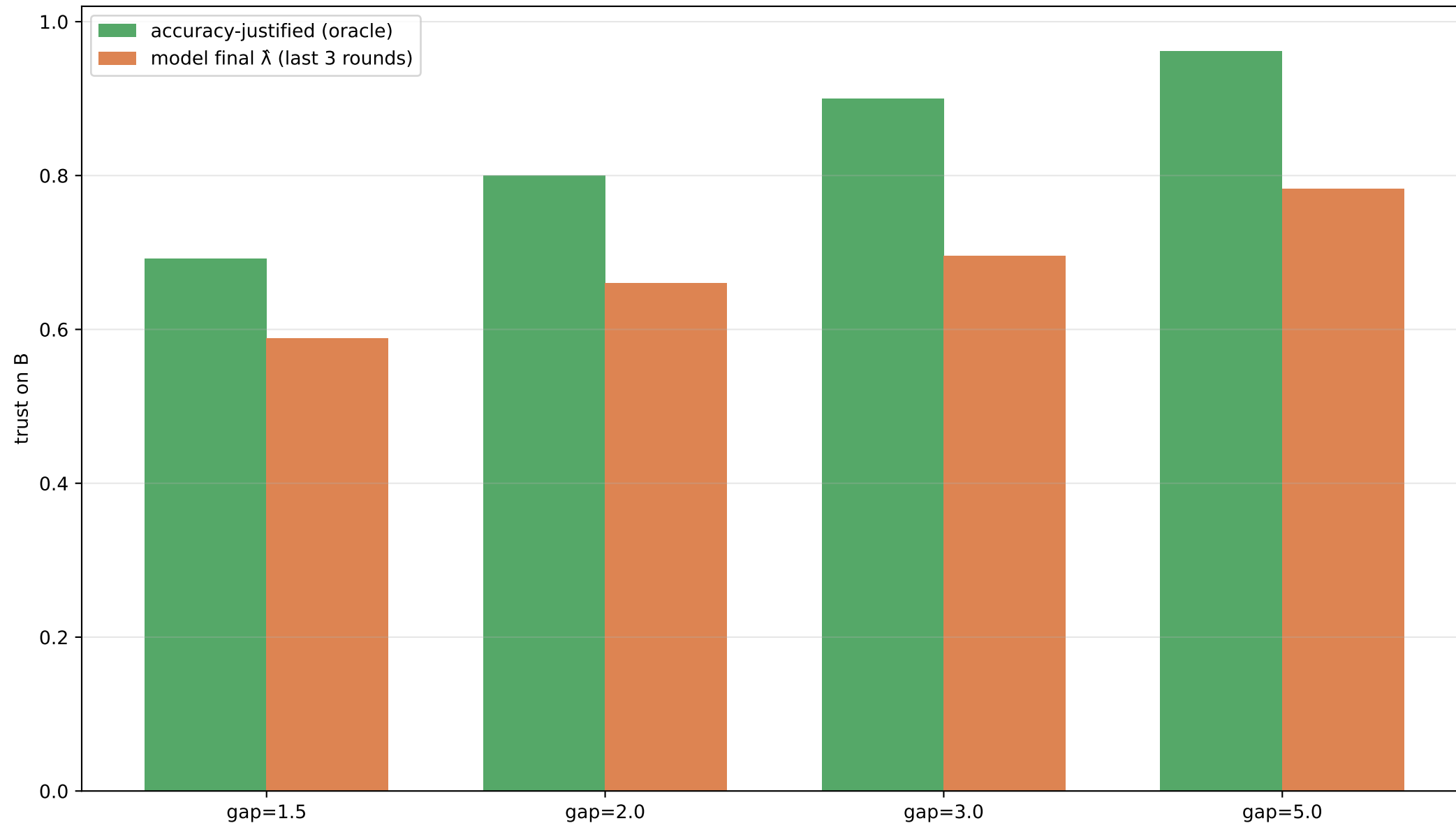
THE key plot: as the accuracy gap widens (easier to tell sources apart) both flip sooner. Taller model bars than Bayes = the model needs more evidence to overcome reputation.

Reputation-STRENGTH effect (gap=2.0): how persistent authority slows the A→B flip

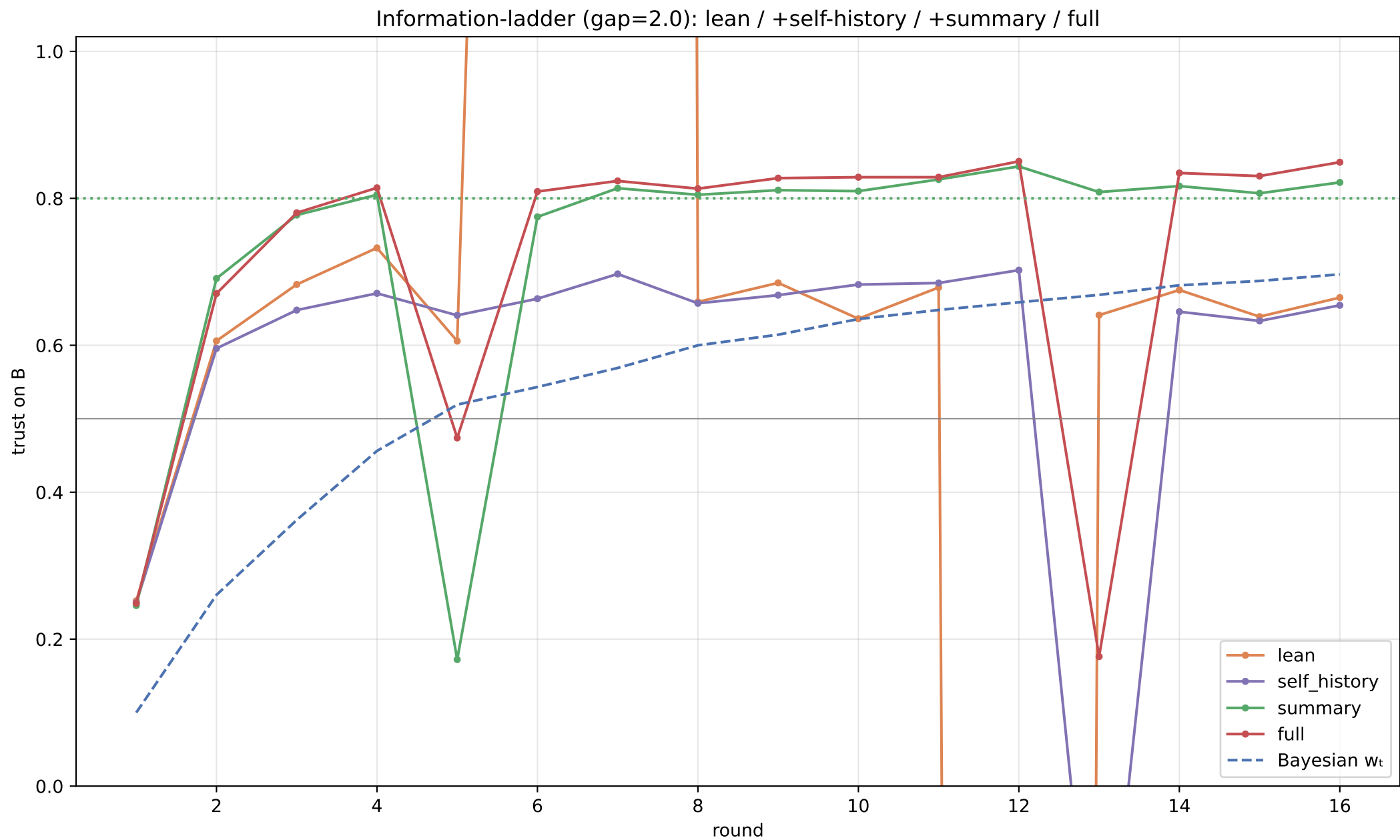


More persistently asserted reputation (faint→moderate→strong) should push the model's trajectory lower / later.

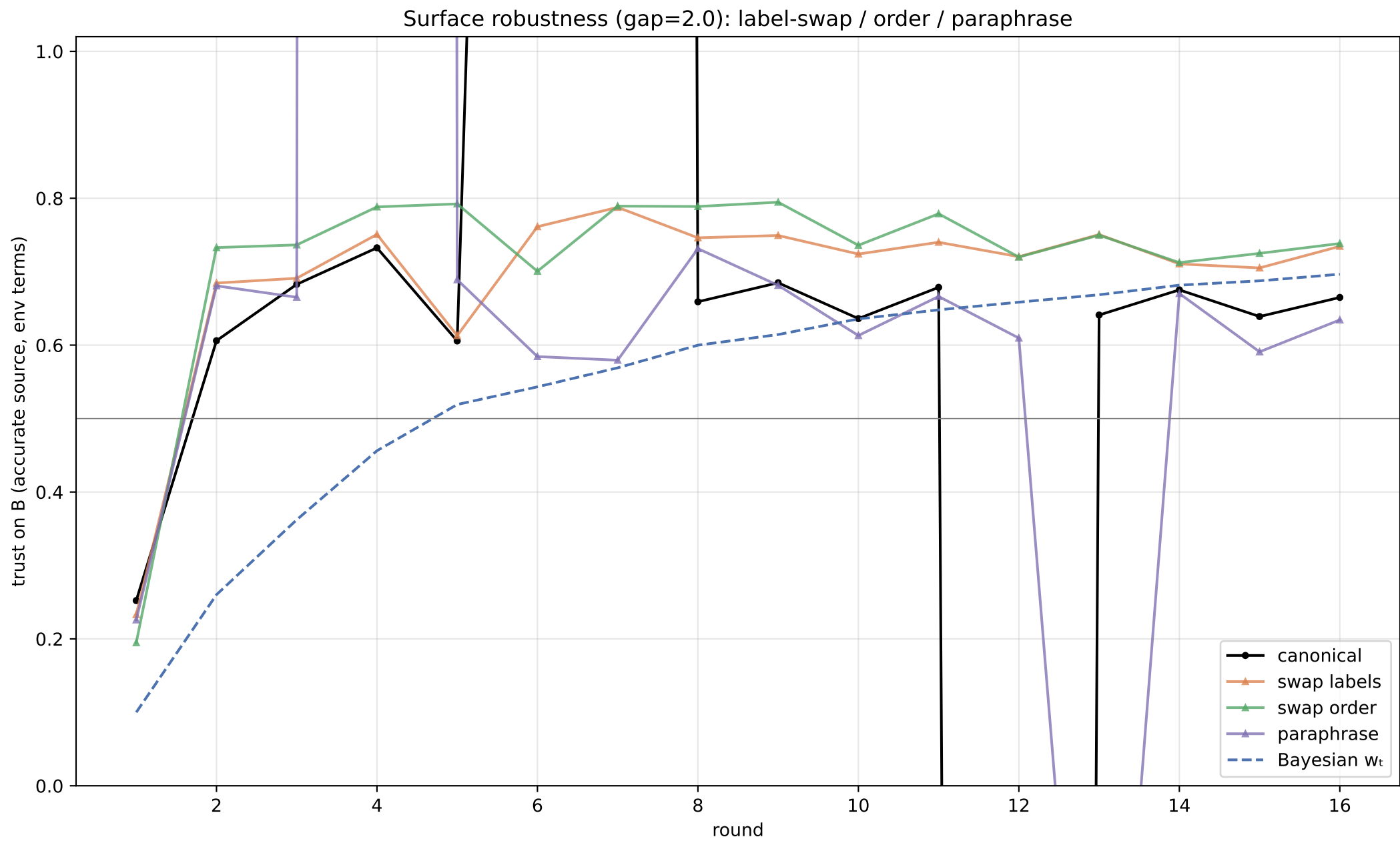
Asymptote — final model trust vs. the accuracy-justified weight



Gap between green (justified) and orange (model) at the end = residual deference to reputation that never resolves.



Localizes the bottleneck: if +summary (handed reliability) beats lean, it's an INFERENCE limit; if even full fails, it's DEFERENCE; if +self-history shifts it, anchoring.



Trajectories should overlap — the A→B migration shouldn't depend on which letter is reputable, the listing order, or wording.

Summary — flip round (model vs Bayes), lag, final trust, verdict

condition	gap	rep	info	flip model	flip Bayes	lag	final $\bar{\lambda}$	oracle	verdict
gap1.5_moderate_le	1.5	moderate	lean	2	10	-8	0.59	0.69	learns→B
gap2.0_moderate_le	2.0	moderate	lean	2	5	-3	0.66	0.80	learns→B
gap3.0_moderate_le	3.0	moderate	lean	2	3	-1	0.70	0.90	learns→B
gap5.0_moderate_le	5.0	moderate	lean	2	2	+0	0.78	0.96	learns→B
gap2.0_faint_lean	2.0	faint	lean	2	5	-3	0.65	0.80	learns→B
gap2.0_strong_lean	2.0	strong	lean	2	5	-3	0.56	0.80	learns→B
gap2.0_moderate_self_h	2.0	moderate	self_history	2	5	-3	0.64	0.80	learns→B
gap2.0_moderate_summ	2.0	moderate	summary	2	5	-3	0.82	0.80	learns→B
gap2.0_moderate_full	2.0	moderate	full	2	5	-3	0.84	0.80	learns→B
gap2.0_robust_swapla	2.0	moderate	lean	2	5	-3	0.72	0.80	learns→B
gap2.0_robust_orde	2.0	moderate	lean	2	5	-3	0.73	0.80	learns→B
gap2.0_robust_paraph	2.0	moderate	lean	2	5	-3	0.63	0.80	learns→B
gap2.0_moderate_no_ad	2.0	moderate	no_advisor	-	-	-	-	-	RMSE=142 (necessity)

lag = model flip round minus Bayesian flip round (positive = behind). final $\bar{\lambda}$ vs oracle shows residual deference.

Takeaways

- Does it learn to trust the accurate source? YES — at gap=2.0 the model's revealed trust crosses 0.5 at round 2 (Bayes: 5).
- Reputation-stickiness: starts anchored at $\hat{\lambda}_1=0.25$ (prior favors A) and over-trusts A by -0.93 on average vs. the rational curve; flip lag = -3 rounds.
- Where the transition sits vs. Bayes across gaps (model vs Bayes flip round): gap1.5:2/10, gap2.0:2/5, gap3.0:2/3, gap5.0:2/2.
- Asymptote: final $\hat{\lambda}=0.66$ vs accuracy-justified 0.80 — residual deference remains.
- Inference vs. deference: compare lean vs. +summary on S8 — if handing the model running accuracy helps, the bottleneck is inference; if not, it's deference.
- Sanity: no-advisor RMSE=142 (advisors must add info); comprehension correct in 100% of probes.
- Caveats: scoped run (~20 games/condition → wider CIs than the 1000-game target); thinking disabled; single model; synthetic Gaussian world.